XU LUO

University of Electronic Science and Technology of China
https://frankluox.github.io

EDUCATION

- **University of Electronic Science and Technology of China** *Sept. 2022 - Present*
  Ph.D Degree in Computer Science
- **University of Electronic Science and Technology of China** *Sept. 2019 - June 2022*
  Master Degree in Computer Science
- **Xiamen University** *Sept. 2015 - June 2019*
  Bachelor Degree in Mathematics

RESEARCH EXPERIENCE

- **Center of Future Media, UESTC** *June. 2024 - Present*
  Adviser: Jingkuan Song
  Focus Area: Robotics
  Project: 1. Understanding generalization capability of generalist robot policies. 2. Reinforcement learning for improving pretrained robot policies.
- **Shanghai AI Laboratory** *Oct. 2023 - June. 2024*
  Adviser: Peng Gao
  Focus Area: AIGC and Agents
  Project: Deep involvement in the project of Lumina T2X series of models that generate any modality with text inputs; designing autonomous mobile phone agents powered by multimodal models.
- **Center of Future Media, UESTC** *Sept. 2022 - Oct. 2023*
  Adviser: Jingkuan Song
  Focus Area: Few-shot learning
  Project: actualizing the potential of foundation models and shaping them into autonomous agents capable of performing intricate tasks in real-world scenarios.
- **Statistical Machine Intelligence Learning Laboratory, UESTC** *Sept. 2019 - June 2022*
  Adviser: Zenglin Xu
  Focus Area: Few-shot transfer, Pretrained models, Representation learning
  Project: Responsible for understanding the key difficulties of few-shot adaptation of pre-trained models towards downstream tasks and exploring its relationship with visual representation learning.
- **Huawei Cloud AI Lab, Beijing** *Mar 2021 - Aug. 2021*
  Adviser: Longhui Wei
  Focus Area: Few-shot learning, Contrastive learning
  Project: Responsible for exploring the negative effect of image background on few-shot learning and designing a feasible solution.

PUBLICATIONS

- **Xu Luo**, Jiaying Yang, Hao Wu, Junlin Xie, Zijin Hong, Ji Zhang, Lianli Gao, Jingkuan Song. ExRL: Sample-Efficient Online Adaptation of Long-Chunk Policies via Execution Length Control. Preprint, 2026.

- Hao Wu*, **Xu Luo***, Zijin Hong, Ji Zhang, Lianli Gao, Jingkuan Song. Pushing Your Robot Policy toward Its Speed Limits via Coupled Gain–Duration Optimization. Preprint, 2026.
- Youguang Xing*, **Xu Luo***, Lianli Gao, Hengtao Shen, Jingkuan Song. Shortcut Learning in Generalist Robot Policies: The Role of Dataset Diversity and Fragmentation. CoRL 2025.
- **Xu Luo***, Jiaying Yang*, Junlin Xie, Ji Zhang, Lianli Gao, Jingkuan Song. Test-Time Scaling of Vision-Language-Action Models via Self-Certainty. CoRL 2025 Eval-Deploy Workshop.
- Junlin Xie*, **Xu Luo***, Hao Wu*, Youguang Xing, Ji Zhang, Lianli Gao, Jingkuan Song. In-Context Adaptation for Generalizable Imitation Learning. CoRL 2025 RemembeRL Workshop (Spotlight).
- Shihan Wu*, **Xu Luo***, Ji Zhang, Junlin Xie, Lianli Gao, Jingkuan Song. Policy Contrastive Decoding for Robotic Foundation Models. ICLR 2026.
- Peng Gao*, Le Zhuo*, Dongyang Liu*, Ruoyi Du*, **Xu Luo***, Longtian Qiu*, Yuhang Zhang, et al. Lumina-T2X: Transforming Text into Any Modality, Resolution, and Duration via Flow-based Large Diffusion Transformers. ICLR 2025.
- Le Zhuo, Ruoyi Du, Han Xiao,..., **Xu Luo**, et al. Lumina-Next: Making Lumina-T2X Stronger and Faster with Next-DiT. NeurIPS 2024.
- Cheng Chen, Junchen Zhu, **Xu Luo**, Hengtao Shen, Lianli Gao, Jingkuan Song. CoIN: A Benchmark of Continual Instruction tuning for Multimodel Large Language Model. NeurIPS 2024.
- Ji Zhang, **Xu Luo**, Lianli Gao, Difan Zou, Heng Tao Shen, Jingkuan Song. From Channel Bias to Feature Redundancy: Uncovering the "Less is More" Principle in Few-Shot Learning. TPAMI 2026.
- Ji Zhang, Lianli Gao, **Xu Luo**, Hengtao Shen, Jingkuan Song. DETA: Denoised Task Adaptation for Few-shot Learning. ICCV 2023.
- **Xu Luo***, Hao Wu*, Ji Zhang, Lianli Gao, Jing Xu and Jingkuan Song. A Closer Look at Few-shot Classification Again. ICML 2023.
- Jing Xu, **Xu Luo**, Xinglin Pan, Yanan Li, Wenjie Pei and Zenglin Xu. Alleviating the sample selection bias in few-shot learning by removing projection to the centroid. NeurIPS 2022.
- **Xu Luo**, Jing Xu and Zenglin Xu. Channel importance matters in few-shot image classification. ICML 2022.
- **Xu Luo**, Longhui Wei, Liangjian Wen, Jinrong Yang, Lingxi Xie, Zenglin Xu, and Qi Tian. Rectifying the shortcut learning of background for few-shot learning. NeurIPS 2021.

AWARDS

- Editorial Board Outstanding Reviewer (5/8400) of ICRA 2026.
- Gold Reviewer of ICML 2026 (Top 25%).